Exp 20

Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Smart Toll Payment System</title>

<!-- QR code library -->
<script
src="https://cdnjs.cloudflare.com/ajax/libs/qrcodejs/1.0.0/qrcode.min.js"></script>

<style>
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
  font-family: Arial, sans-serif;
}
```

```
body {
  background: #f2f5f9;
  color: #263238;
}

header {
  background: #174a7e;
  color: white;
  text-align: center;
  padding: 24px;
}

header p {
  margin-top: 8px;
}

nav {
  background: #0d3155;
  padding: 15px;
  text-align: center;
}

nav a {
  color: white;
  text-decoration: none;
```

```
    margin: 0 12px;
}

nav a:hover {
    color: #a9d7ff;
}

.hero {
    background: #dcecff;
    text-align: center;
    padding: 40px 20px;
}

.hero h2 {
    color: #174a7e;
    margin-bottom: 12px;
}

.container {
    max-width: 1100px;
    margin: auto;
    padding: 25px 16px;
}

.section-title {
    color: #174a7e;
```

```
  margin-bottom: 18px;
}

.cards {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(220px, 1fr));
  gap: 20px;
}

.card, .panel {
  background: white;
  border-radius: 10px;
  padding: 22px;
  box-shadow: 0 3px 12px #00000010;
}

.card h3 {
  color: #174a7e;
  margin-bottom: 12px;
}

.card p {
  margin: 10px 0;
}

.panel {
```

```
    margin-top: 25px;
}

button {
    background: #176bc1;
    color: white;
    border: none;
    border-radius: 5px;
    padding: 11px 16px;
    cursor: pointer;
    margin-top: 8px;
}

button:hover {
    background: #0d477f;
}

input, select {
    width: 100%;
    padding: 12px;
    margin: 8px 0 15px;
    border: 1px solid #cbd3dc;
    border-radius: 5px;
}

label {
```

```
  font-weight: bold;
}

.balance {
  background: #e1f5e9;
  padding: 20px;
  border-radius: 8px;
  margin-bottom: 18px;
}

.balance h3 {
  color: #176b42;
}

#qrCode {
  margin: 20px 0;
}

#qrCode img, #qrCode canvas {
  margin: auto;
}

#message, #passMessage, #verifyResult {
  margin-top: 12px;
  font-weight: bold;
  overflow-wrap: anywhere;
```

```
}

footer {
  background: #0d3155;
  color: white;
  text-align: center;
  padding: 20px;
  margin-top: 30px;
}

@media(max-width: 600px) {
  nav a {
    display: inline-block;
    margin: 7px;
  }
}
</style>
</head>

<body>

<header>
  <h1> 🚘 Smart Toll Payment System</h1>
  <p>Fast, Secure and Cashless Toll Payments</p>
</header>
```

```
<nav>
  <a href="#home">Home</a>
  <a href="#passes">Toll Passes</a>
  <a href="#advance">Advance Payment</a>
  <a href="#wallet">E-Wallet</a>
  <a href="#verify">Verify Transaction</a>
</nav>

<section class="hero" id="home">
  <h2>Travel Without Unnecessary Delays</h2>
  <p>Purchase toll passes, manage your wallet and generate QR codes.</p>
  <button onclick="goToPasses()">Explore Toll Passes</button>
</section>

<main class="container">

  <!-- Toll passes -->
  <section id="passes">
    <h2 class="section-title">Available Toll Passes</h2>

    <div class="cards">

      <div class="card">
        <h3>Single Journey</h3>
        <p>For one toll crossing.</p>
        <p>Validity: One journey</p>
```

```
    <h3>₹100</h3>
    <button onclick="buyPass('Single Journey', 100)">
      Purchase Pass
    </button>
  </div>

  <div class="card">
    <h3>Return Journey</h3>
    <p>For onward and return travel.</p>
    <p>Validity: Same-day demo</p>
    <h3>₹150</h3>
    <button onclick="buyPass('Return Journey', 150)">
      Purchase Pass
    </button>
  </div>

  <div class="card">
    <h3>Monthly Pass</h3>
    <p>For regular toll users.</p>
    <p>Validity: 30 days in this demo</p>
    <h3>₹1500</h3>
    <button onclick="buyPass('Monthly Pass', 1500)">
      Purchase Pass
    </button>
  </div>
```

```
  </div>

  <div class="panel">
    <h3 class="section-title">Vehicle Details</h3>

    <label>Vehicle Registration Number</label>
    <input type="text" id="vehicleNumber"
        placeholder="Example: AP 01 AB 1234">

    <label>Vehicle Type</label>
    <select id="vehicleType">
      <option>Car / Jeep / Van</option>
      <option>Light Commercial Vehicle</option>
      <option>Bus / Truck</option>
      <option>Heavy Vehicle</option>
    </select>

    <label>Selected Pass</label>
    <input type="text" id="selectedPass" readonly
        placeholder="Select a pass above">

    <button onclick="confirmPass()">Confirm Pass</button>
    <p id="passMessage"></p>
  </div>
</section>
```

```
<!-- Advance payment -->
<section class="panel" id="advance">
  <h2 class="section-title">Advance Toll Payment</h2>

  <label>Vehicle Registration Number</label>
  <input type="text" id="advanceVehicle"
      placeholder="Enter vehicle number">

  <label>Toll Plaza</label>
  <select id="tollPlaza">
    <option value="Highway Toll Plaza">Highway Toll Plaza</option>
    <option value="City Bypass Toll Plaza">City Bypass Toll Plaza</option>
    <option value="Expressway Toll Plaza">Expressway Toll Plaza</option>
  </select>

  <label>Journey Date</label>
  <input type="date" id="journeyDate">

  <label>Advance Amount (₹)</label>
  <input type="number" id="advanceAmount"
      min="1" placeholder="Enter amount">

  <button onclick="makeAdvancePayment()">
    Pay in Demo
  </button>
```

```
  <p id="message"></p>
</section>

<!-- Wallet -->
<section class="panel" id="wallet">
  <h2 class="section-title">💳 E-Wallet Management</h2>

  <div class="balance">
    <h3>Available Balance</h3>
    <h2 id="walletBalance">₹500.00</h2>
  </div>

  <label>Add Money (₹)</label>
  <input type="number" id="addAmount"
      min="1" placeholder="Enter amount">

  <button onclick="addMoney()">Add Money (Demo)</button>

  <br><br>

  <label>Pay Toll from Wallet (₹)</label>
  <input type="number" id="deductAmount"
      min="1" placeholder="Enter toll amount">

  <button onclick="payFromWallet()">Pay Toll</button>
```

```
  <p id="walletMessage"></p>
</section>

<!-- QR code -->
<section class="panel" id="qrSection">
  <h2 class="section-title">🧮 Toll QR Code</h2>
  <p>Generate a QR code for a pass or advance payment.</p>

  <button onclick="generateQR()">Generate QR Code</button>

  <div id="qrCode"></div>
  <p id="qrMessage"></p>
</section>

<!-- Transaction verification -->
<section class="panel" id="verify">
  <h2 class="section-title">🔎 Verify Toll Transaction</h2>

  <label>Transaction ID</label>
  <input type="text" id="transactionId"
      placeholder="Enter transaction ID">

  <button onclick="verifyTransaction()">
    Verify Transaction
  </button>
```

```
    <p id="verifyResult"></p>
  </section>

  <!-- History -->
  <section class="panel" id="history">
    <h2 class="section-title">Transaction History</h2>
    <div id="transactionHistory">No transactions yet.</div>
  </section>

</main>

<footer>
  <p>© 2026 Smart Toll Payment System</p>
  <p>Academic Project Demonstration</p>
</footer>

<script>
let wallet = 500;
let passName = "";
let passPrice = 0;
let currentPass = null;
let transactions = [];
let counter = 1001;
let lastTransaction = null;

document.getElementById("journeyDate").min =
```

```
    new Date().toLocaleDateString("en-CA");

function goToPasses() {
    document.getElementById("passes").scrollIntoView({
        behavior: "smooth"
    });
}

function buyPass(name, price) {
    passName = name;
    passPrice = price;

    document.getElementById("selectedPass").value =
        name + " - ₹" + price;

    document.getElementById("passMessage").textContent =
        "Selected " + name + ". Enter vehicle details and confirm.";
}

function createTransaction(type, amount, details) {
    const transaction = {
        id: "TOLL" + counter++,
        type: type,
        amount: amount,
        details: details,
        date: new Date().toLocaleString(),
```

```
    status: "Demo Payment Recorded"
  };

  transactions.unshift(transaction);
  lastTransaction = transaction;

  renderHistory();
  return transaction;
}

function confirmPass() {
  const vehicle = document.getElementById("vehicleNumber")
    .value.trim().toUpperCase();

  const vehicleType = document.getElementById("vehicleType").value;

  if (!passName || !passPrice) {
    document.getElementById("passMessage").textContent =
      "Please select a toll pass first.";
    return;
  }

  if (!/^[A-Z0-9 -]{5,15}$/.test(vehicle)) {
    document.getElementById("passMessage").textContent =
      "Enter a valid vehicle registration number.";
    return;
```

```
}

if (wallet < passPrice) {
  document.getElementById("passMessage").textContent =
    "Insufficient wallet balance. Add money first.";
  return;
}

wallet -= passPrice;
currentPass = {
  name: passName,
  price: passPrice,
  vehicle: vehicle,
  vehicleType: vehicleType
};

const transaction = createTransaction(
  "Toll Pass Purchase",
  passPrice,
  passName + " | " + vehicle
);

updateWallet();

document.getElementById("passMessage").textContent =
  "Demo pass purchase recorded. Transaction ID: " +
```

```
      transaction.id + ". Amount: ₹" + passPrice +
      ". No real toll pass has been issued.";
}

function makeAdvancePayment() {
   const vehicle = document.getElementById("advanceVehicle")
      .value.trim().toUpperCase();

   const plaza = document.getElementById("tollPlaza").value;
   const date = document.getElementById("journeyDate").value;
   const amount = Number(document.getElementById("advanceAmount").value);

   const today = new Date();
   today.setHours(0, 0, 0, 0);

   if (!/^[A-Z0-9 -]{5,15}$/.test(vehicle)) {
      document.getElementById("message").textContent =
         "Enter a valid vehicle registration number.";
      return;
   }

   if (!date || new Date(date + "T00:00:00") < today) {
      document.getElementById("message").textContent =
         "Select today or a future journey date.";
      return;
   }
```

```
if (!Number.isFinite(amount) || amount <= 0) {
  document.getElementById("message").textContent =
    "Enter a valid payment amount.";
  return;
}

if (wallet < amount) {
  document.getElementById("message").textContent =
    "Insufficient wallet balance. Add money first.";
  return;
}

wallet -= amount;

const transaction = createTransaction(
  "Advance Toll Payment",
  amount,
  vehicle + " | " + plaza + " | " + date
);

updateWallet();

document.getElementById("message").textContent =
  "Demo advance payment recorded. Transaction ID: " +
  transaction.id + ". No real payment was processed.";
```

```
}

function addMoney() {
  const amount = Number(document.getElementById("addAmount").value);

  if (!Number.isFinite(amount) || amount <= 0) {
    document.getElementById("walletMessage").textContent =
      "Enter a valid amount.";
    return;
  }

  if (amount > 100000) {
    document.getElementById("walletMessage").textContent =
      "Demo top-up limit is ₹100,000.";
    return;
  }

  wallet += amount;

  const transaction = createTransaction(
    "Demo Wallet Top-up",
    amount,
    "Wallet recharge"
  );

  updateWallet();
```

```
  document.getElementById("walletMessage").textContent =
    "Demo top-up recorded. Transaction ID: " + transaction.id +
    ". No real money was added.";

  document.getElementById("addAmount").value = "";
}

function payFromWallet() {
  const amount = Number(document.getElementById("deductAmount").value);

  if (!Number.isFinite(amount) || amount <= 0) {
    document.getElementById("walletMessage").textContent =
      "Enter a valid toll amount.";
    return;
  }

  if (amount > wallet) {
    document.getElementById("walletMessage").textContent =
      "Insufficient wallet balance.";
    return;
  }

  wallet -= amount;

  const transaction = createTransaction(
```

```
      "Toll Payment",
      amount,
      "Wallet toll payment"
    );

    updateWallet();

    document.getElementById("walletMessage").textContent =
      "Demo toll payment recorded. Transaction ID: " +
      transaction.id + ". No real toll was paid.";

    document.getElementById("deductAmount").value = "";
}

function updateWallet() {
    document.getElementById("walletBalance").textContent =
      "₹" + wallet.toFixed(2);
}

function generateQR() {
    const box = document.getElementById("qrCode");
    const message = document.getElementById("qrMessage");

    box.replaceChildren();

    if (!lastTransaction) {
```

```
    message.textContent =
      "Make a demo transaction first to generate its QR code.";
    return;
}

if (typeof QRCode === "undefined") {
    message.textContent =
      "QR library could not load. Check your internet connection.";
    return;
}

const qrData = JSON.stringify({
    transactionId: lastTransaction.id,
    type: lastTransaction.type,
    amount: lastTransaction.amount,
    status: lastTransaction.status
});

new QRCode(box, {
    text: qrData,
    width: 180,
    height: 180
});

message.textContent =
    "Demo QR code generated for transaction " +
```

```
      lastTransaction.id +
      ". It is not an official toll authorization.";
}

function verifyTransaction() {
  const id = document.getElementById("transactionId")
    .value.trim().toUpperCase();

  const transaction = transactions.find(item => item.id === id);
  const output = document.getElementById("verifyResult");

  if (!transaction) {
    output.textContent =
      "Transaction not found in this browser session.";
    output.style.color = "#b3261e";
    return;
  }

  output.style.color = "#176b42";
  output.textContent =
    "Demo record found | ID: " + transaction.id +
    " | Type: " + transaction.type +
    " | Amount: ₹" + transaction.amount +
    " | Details: " + transaction.details +
    " | Status: " + transaction.status;
}
```

```
function renderHistory() {
  const container = document.getElementById("transactionHistory");
  container.replaceChildren();

  if (transactions.length === 0) {
    container.textContent = "No transactions yet.";
    return;
  }

  transactions.forEach(transaction => {
    const row = document.createElement("div");
    row.style.padding = "12px 0";
    row.style.borderBottom = "1px solid #eee";

    const details = document.createElement("p");
    details.textContent =
      transaction.id + " | " + transaction.type +
      " | ₹" + transaction.amount +
      " | " + transaction.date;

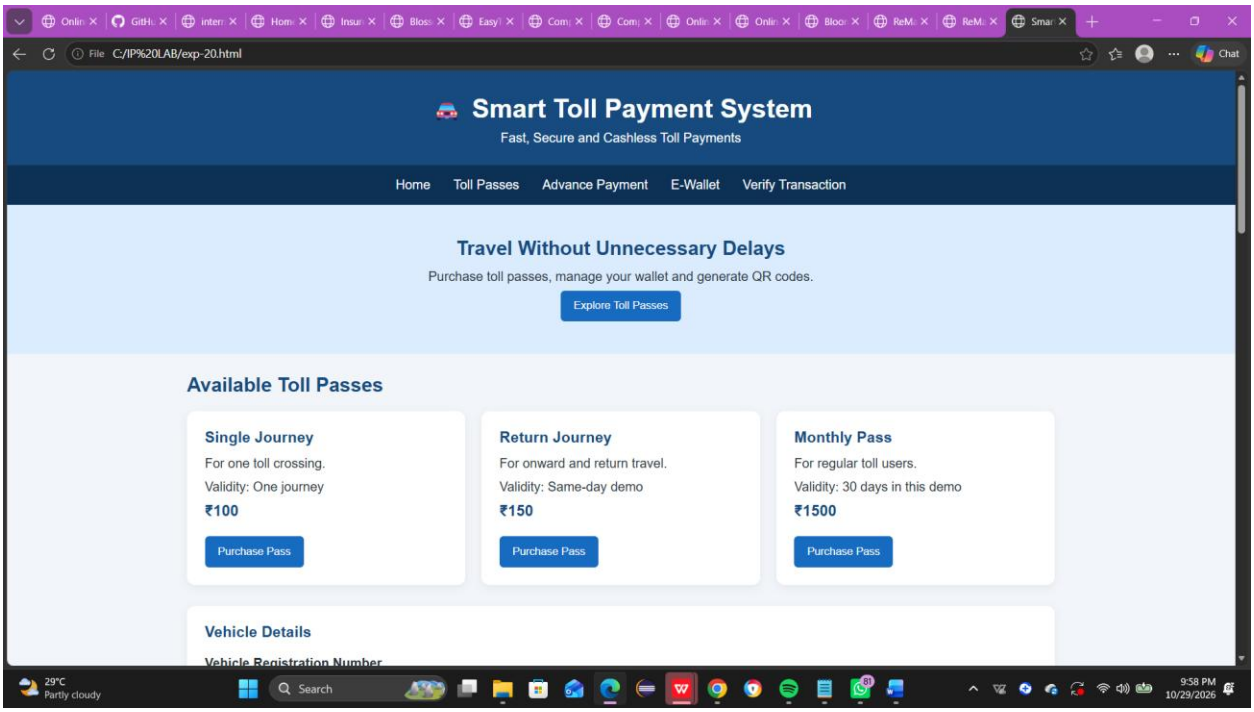
    const verifyButton = document.createElement("button");
    verifyButton.textContent = "Verify";
    verifyButton.onclick = function () {
      document.getElementById("transactionId").value =
        transaction.id;
```

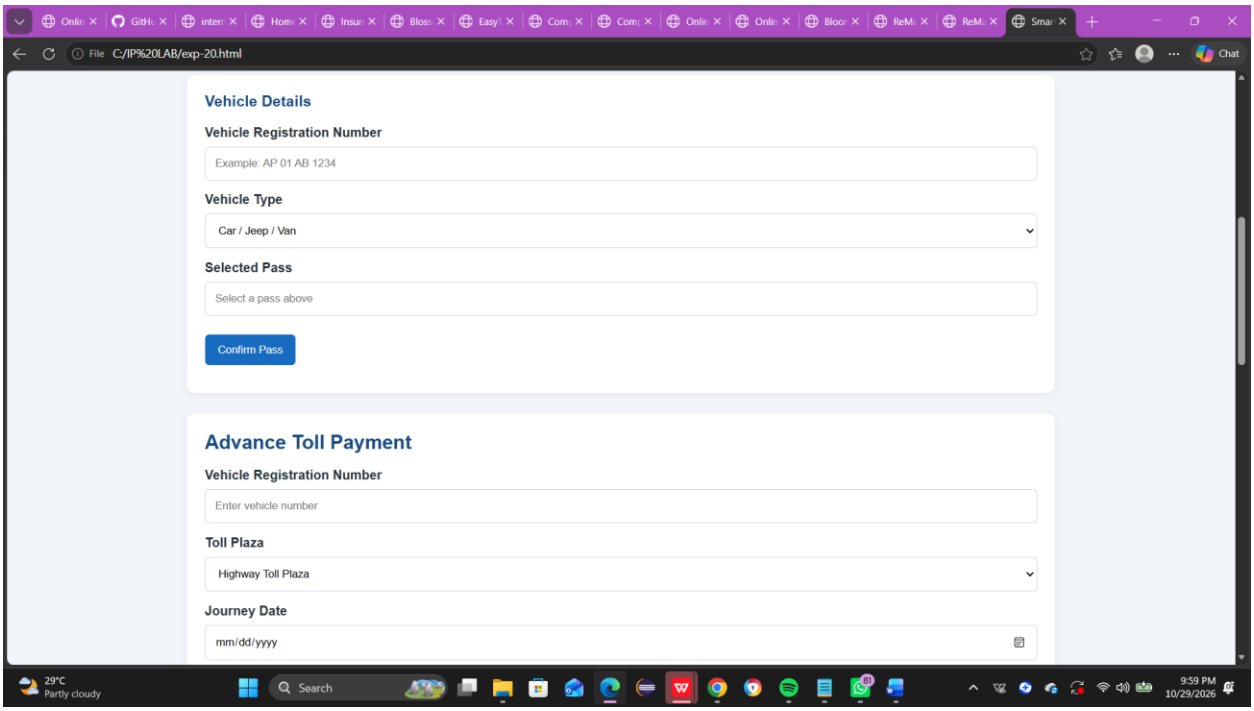
```
        verifyTransaction();
        document.getElementById("verify").scrollIntoView({
          behavior: "smooth"
        });
      };

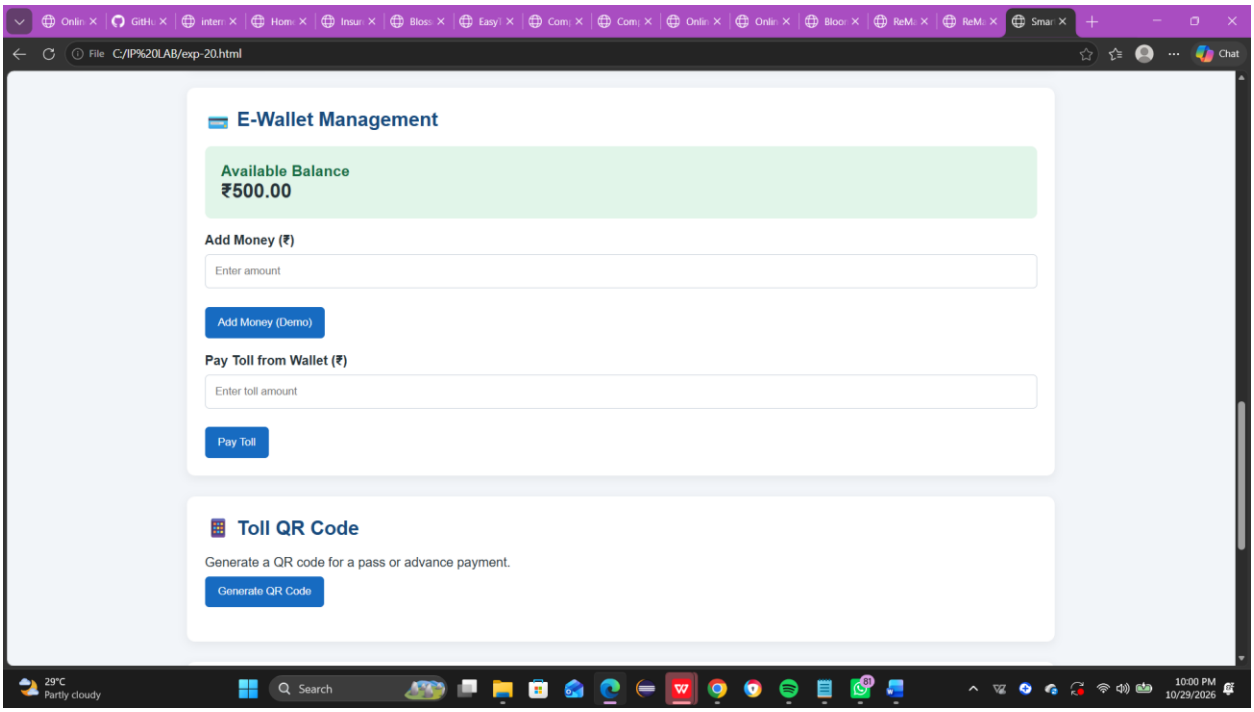
      row.append(details, verifyButton);
      container.appendChild(row);
    });
}
</script>

</body>
</html>
```

Output:


Smart Toll Payment System
Fast, Secure and Cashless Toll Payments
Home
Toll Passes
Advance Payment
E-Wallet
Verify Transaction
Travel Without Unnecessary Delays
Purchase toll passes, manage your wallet and generate QR codes.
Explore Toll Passes
Available Toll Passes
Single Journey
For one toll crossing.
Validity: One journey
₹100
Purchase Pass
Return Journey
For onward and return travel.
Validity: Same-day demo
₹150
Purchase Pass
Monthly Pass
For regular toll users.
Validity: 30 days in this demo
₹1500
Purchase Pass
Vehicle Details
Vehicle Registration Number


File C:/IP%20LAB/exp-20.html
Vehicle Details
Vehicle Registration Number
Example: AP 01 AB 1234
Vehicle Type
Car / Jeep / Van
Selected Pass
Select a pass above
Confirm Pass
Advance Toll Payment
Vehicle Registration Number
Enter vehicle number
Toll Plaza
Highway Toll Plaza
Journey Date
mm/dd/yyyy
29°C
Partly cloudy
Search
9:59 PM
10/29/2026


File C:/IP%20LAB/exp-20.html
E-Wallet Management
Available Balance
₹500.00
Add Money (₹)
Enter amount
Add Money (Demo)
Pay Toll from Wallet (₹)
Enter toll amount
Pay Toll
Toll QR Code
Generate a QR code for a pass or advance payment.
Generate QR Code
29°C
Partly cloudy
Search
10:00 PM
10/29/2026